From social media to clinical insight: contrastive learning on families' narratives in an ultra-rare developmental and epileptic encephalopathy

Emma Le Priol[1,@], Mélissa Cassard [2], Laetitia Gherardhi [2], Rima Nabbout [3], Anita Burgun[1,4]

1. Clinical Bioinformatics Laboratory, Université Paris Cité, INSERM UMR 1163, Imagine Institute, Paris, France
2. *KCNB1 France* Association, 6 rue des Martins, 44230 Saint-Sébastien -sur-Loire, France
3. Reference Center for Rare Epilepsies, Department of Pediatric Neurology, Necker-Enfants Malades University Hospital, APHP, Member of European Reference Network EpiCARE, Université Paris Cité, INSERM UMR 1163, Imagine Institute, Paris, France
4. Department of Medical informatics, Necker-Enfants Malades University Hospital, APHP, Paris, France

@ Corresponding author: emmalepriol@gmail.com

BACKGROUND

A mutation in the KCNB1 gene causes developmental and epileptic encephalopathy (DEE), an ultra-rare disease characterised by **frequent seizures** of various types, as well as **general developmental delay**, or even **regression or stagnation in developmental milestones**. Families affected by these conditions, as with many rare diseases, face **numerous challenges**, including diagnostic uncertainty, as well as social and family difficulties, a sense of isolation when dealing with a very rare condition, and the lack of treatment, …

For all these reasons, these families may **turn to online forums or social media to share their experiences, questions and concerns**. This real-life data provides valuable information about how families experience the condition and can help to expand our understanding of it, ultimately leading to even more tailored care.

OBJECTIVES

(i) Producing less anisotropic, more informative and clusterable document embeddings for social media narratives written in French and about an ultra-rare disease.

(ii) Deepening the understanding of KCNB1-DEE.

DATA

Working with the *KCNB1 France* association, we were able to retrieve and analyse messages exchanged by families between August 2021 and June 2026 in closed groups (Messenger, Facebook and Whatsapp) dedicated to families. These messages have been pseudonymised and are stored securely. The processing of this data is carried out in accordance with reference methodology MR-004 and is registered with the HDH under number F20231205165541.

METHODS

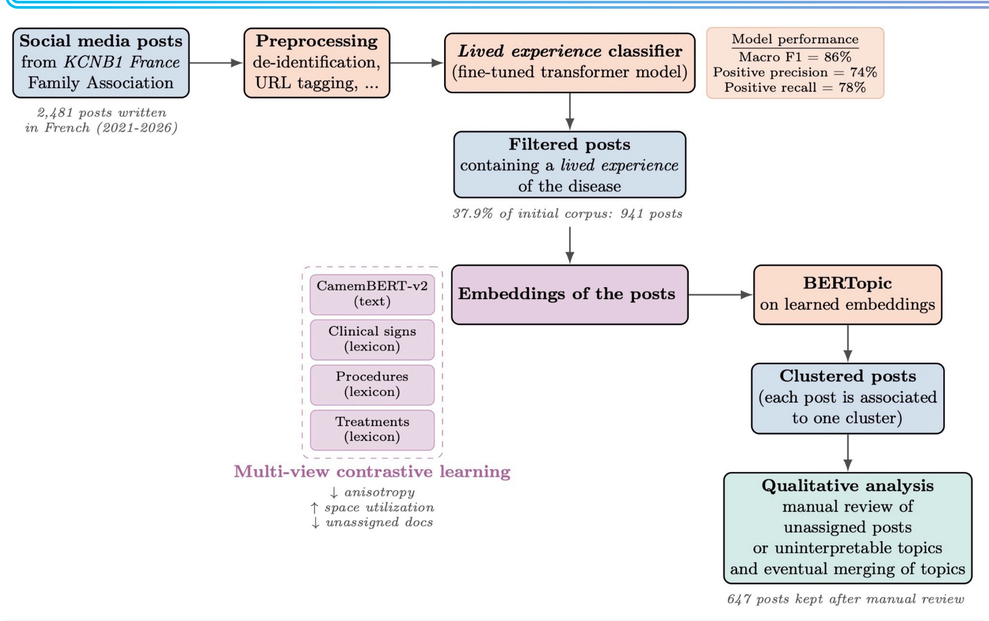


Figure 1 – methodological pipeline

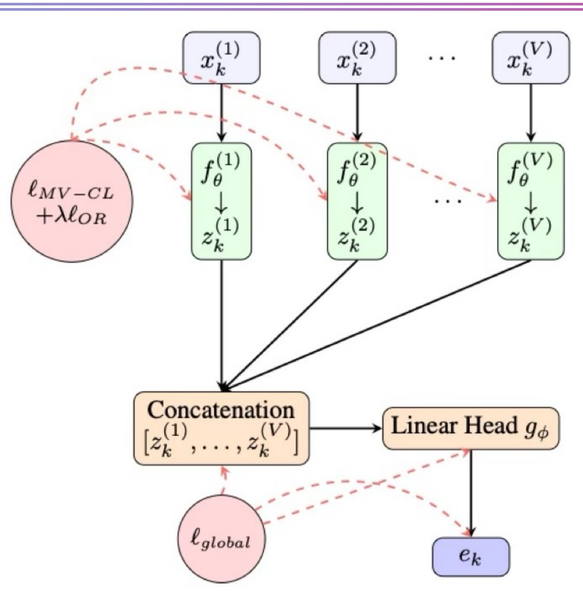


Figure 2 – structure of the contrastive learning framework used to get improved embeddings

Heterogeneous views $x_k^{(v)}$ (lexicon-based) are projected into a shared latent space through view-specific encoders $f_\theta^{(v)}$. The final document embedding e_k is generated by concatenating the individual view projections $z_k^{(v)}$ and applying a single learnable linear projection g_Φ followed by L_2 -normalization. The fusion process is supervised by ℓ_{global} that ensures the final document representation remains consistent with its constituent views.

RESULTS & DISCUSSION

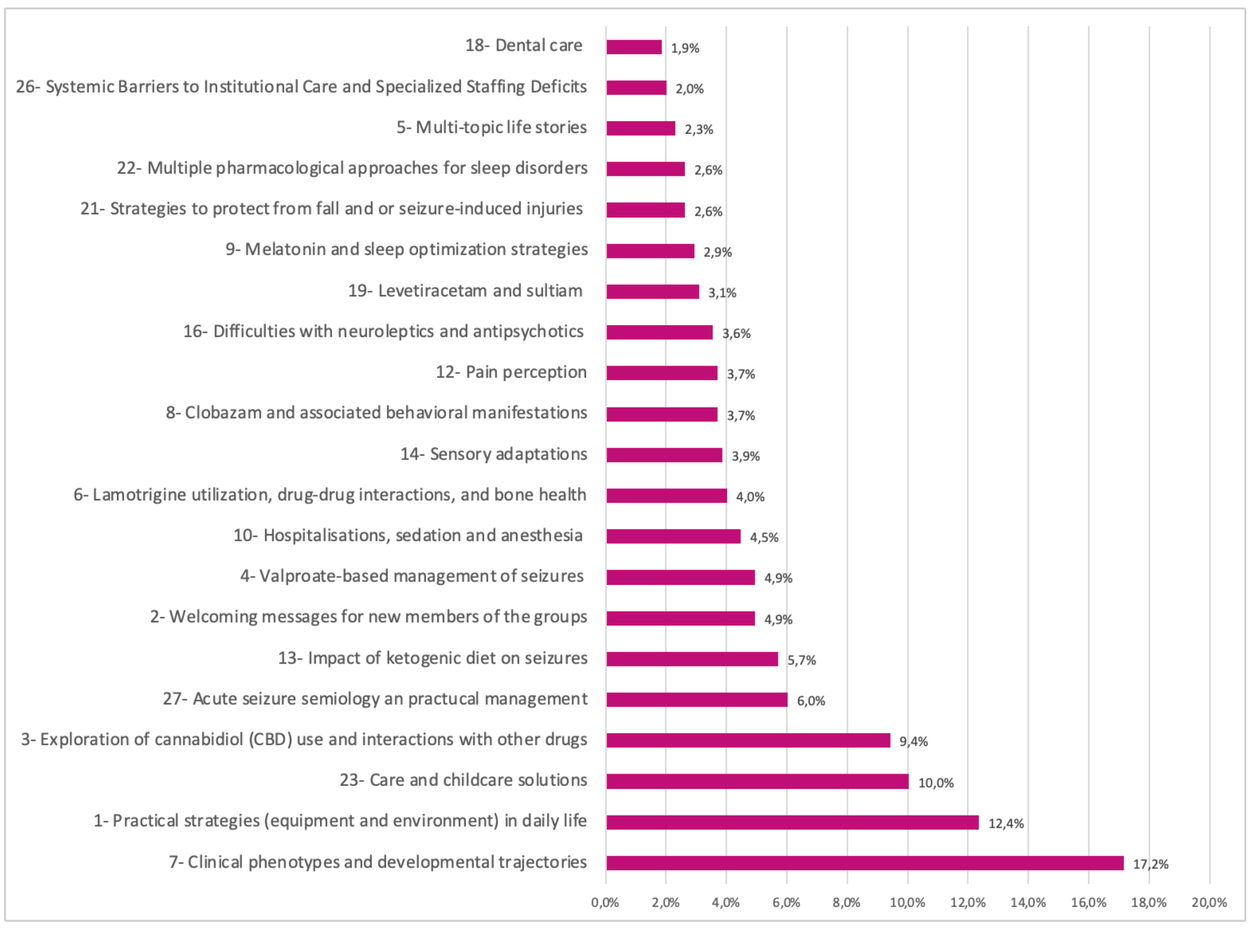


Figure 3 – repartition of the topics (sum >100% because a message can be assigned to multiple topics)

	Text-only	HMVCL $\tau = 0.1$	HMVCL+OR $\tau = 0.1$	HMVCL $\tau = 1$	HMVCL+OR $\tau = 1$
Embedding space geometry (*clustering*)					
Uniformity ↓	∞	$\mathbf{-3.17 \pm 0.04}$	$\underline{-3.03 \pm 0.05}$	-2.39 ± 0.08	-3.02 ± 0.04
Isotropic Coefficient ↑	$8.25 \cdot 10^{-9}$	$\mathbf{0.053 \pm 0.004}$***	$\underline{0.052 \pm 0.002}$***	0.009 ± 0.001***	0.017 ± 0.001***
Avg.cos.sim. ↓	0.88	0.10 ± 0.01***	0.14 ± 0.01***	$\underline{0.05 \pm 0.01}$***	$\mathbf{0.03 \pm 0.01}$***
Effective Rank (Dim)	209 (768)	45 ± 0.3*** (64)	45 ± 0.3*** (64)	$\underline{26 \pm 0.7}$ (64)***	30 ± 0.6 (64)***
Top-10 PCs var. (%)	51.67	59.50 ± 1.06***	58.16 ± 0.96***	96.11 ± 0.33***	92.64 ± 0.98***
Top-1 PCs var. (%)	16.20	7.73 ± 0.22	8.19 ± 0.17	35.20 ± 1.74***	16.45 ± 0.67***
clustering					
Silhouette ↑	$\underline{0.17}$	0.08 ± 0.01	0.10 ± 0.001	$\mathbf{0.26 \pm 0.02}$***	0.12 ± 0.09
Davies-Bouldin ↓	1.79	1.65 ± 0.10***	1.58 ± 0.06*	$\mathbf{1.02 \pm 0.07}$***	$\underline{1.28 \pm 0.09}$***
Pairwise ARI stability across runs ↑	-	0.71 ± 0.38	$\underline{0.72 \pm 0.39}$	$\mathbf{0.81 \pm 0.07}$	0.18 ± 0.16
Outliers Ratio (%) ↓	95.15	12.07 ± 4.78***	$\underline{11.49 \pm 4.49}$***	$\mathbf{10.59 \pm 3.90}$***	34.11 ± 17.85***

*Table 1 – Embedding space & downstream metrics (mean ±std, 15 runs). The best and second best results are highlighted in bold and underlined respectively. For text-only, no std reported as the embeddings are deterministic. *** indicates p<0.001, ** indicates p<0.01 and * indicates p<0.05 for the comparison against the text-only baseline (paired Wilcoxon signed-rank test and Holm correction).*

CLINICAL FINDINGS

FEEDING DIFFICULTIES

- Food aversion, inability to transition to solids, eventual gastrostomy → raises an unresolved question: disease-intrinsic hypotonia vs. caused by ASMs and neuroleptics?
- Parallel with Dravet syndrome literature: feeding difficulties linked to medication introduction/dosage increases; symptom improvement reported after ASM reduction/withdrawal.

SLEEP DISTURBANCES

- Strongly corroborated by a dedicated KCNB1 cohort study (n=78): 67.9% affected.
- Families' narratives complement this cohort study by detailing medication strategies (melatonin dosing, second-line hydroxyzine/risperidone/trazodone) and non-pharmacological approaches.

HETEROGENEITY OF TREATMENT RESPONSE

- Marked inter-individual variability across ASMs: same molecule transformative for some patients, ineffective/poorly tolerated for others; frequent pattern of initial efficacy followed by decline → consistent with the precision-medicine agenda in monogenic epilepsies.
- Similar pattern for the ketogenic diet: strong initial benefit, progressive attenuation, high sensitivity to minor glucose intake, seizure relapse upon discontinuation → consistent with the dietary-therapy literature in drug-resistant pediatric epilepsy.
- Families explore CBD both as antiseizure add-on and as a way to taper off neuroleptics for behavioral comorbidities → consistent with existing CBD-in-epilepsy literature.
- The CBD–clobazam interaction evoked by families → well-characterized interaction
- Divergent reported efficacy of CBD plausibly linked to product type (pharmaceutical-grade vs. unregulated over-the-counter CBD).

SEIZURE SEMIOLOGY AND MODULATING FACTORS

- Reported seizure phenotypes consistent with published cohorts.
- Fever/infection correlation with seizure frequency reported inconsistently: aggravating and protective (both effects are present in the literature on other DEEs, depending on the gene).
- Families also report other seizure-modulating factors (puberty, anesthesia,…) not yet explored in the literature.

SOCIAL/INFORMATIONAL FINDINGS

NEED FOR SEIZURE-RECOGNITION EDUCATION AND CLEAR PROTOCOLS

- Difficulty of the families in recognizing seizures/the type of seizure.
- Seizure-recognition knowledge and acute seizure medication learned largely through peer exchange → the families seem to receive inconsistent instructions from different care teams.
- Gap extends beyond family/clinician : school staff, medico-social carers, emergency responders lack training → rarity implies limited accumulated professional expertise.
- Families spontaneously propose formalizing their knowledge into guides/videos for non-specialist professionals → directly motivates a co-designed information resource.

THE DISPROPORTIONATE BURDEN OF MATERNAL CAREGIVING

- Near-exclusive authorship of narratives by women → visible via French grammatical gender marking.
- Consistent with Dravet syndrome literature (French survey, n=91): mothers sole caregiver in >1/3 of households and >50% interrupted their careers (vs. 7.1% of fathers).

CAREGIVING AND UNCERTAINTY ABOUT THE FUTURE

- Lack of specialized structures and trained staff shifts care burden onto the private sphere → parents become substitute care/education professionals.
- Pediatric-to-adult care transition poorly coordinated, generating anticipatory anxiety about the future → echoing broader intellectual-disability caregiving literature.

CONCLUSION

This study demonstrates that the online spaces built by *KCNB1 France* Family Association for families of patients diagnosed with KCNB1-DEE constitute **far more than a support network**: they are a rich source of caregiver-reported outcomes for a condition which is extremely rare, making traditional data sources scarcer. The hybrid computational-qualitative methodology developed here made it possible to **extract meaningful and interpretable themes** from a corpus presenting several challenging characteristics, and produced observations that both **corroborate** and **extend existing literature on KCNB1-DEE and other DEEs** on a large span of subjects.

Several of the concrete needs voiced by families point directly to **actionable next steps**. We plan to co-design with *KCNB1 France* Family Association an information resource on epilepsy crisis management for medico-social professionals, directly responding to the protocol and training gaps described, and to develop the specifications for a dedicated support application for families, informed by the practical strategies and unmet needs identified in this corpus.

More broadly, this work illustrates, for KCNB1-DEE and for ultra-rare diseases more generally, how social media narratives from families can be captured, and mobilized as a genuine resource for both clinical understanding and public health strategy.

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